

SmRF — Smart Radon Fan

Can I trust SmRF?



Smart Radon Fan

SmRF includes fault-tolerant features

- Goal: SmRF-enabled radon mitigation is as reliable as a standard radon fan installation
- SmRF builds on top of standard radon mitigation systems
- SmRF improves radon mitigation visibility



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SmRF Fault-Tolerance Summary

Failure Type	SmRF Behavior
Airthings sensor failure	Fail active (fan forced ON)
Fan control-path failure	Fail visible (user alerted)
Internet/cloud failure	Local autonomous operation continues
Home Wi-Fi/router failure	Fail visible (user alerted)
Pi IP address change	AP interface reveals new IP Local autonomous operation continues
Shelly IP address change	Automatic rediscovery and reconnection
OTA updates	Signed updates and rollback support
Power interruption	Automatic reboot and recovery



SmRF is designed so that failures degrade toward standard radon fan operation

SmRF Viewer

- Cloud Server that receives regular SmRF site reports
- Site reports include radon levels and installation health
- Missing reports may indicate an installation failure
- Health reports may indicate local sensor failures
- Automatic alerts generated from site reports. E.g.,
 - For N consecutive days: Radon level $> 4.0\text{pCi/L}$ AND Fan ON
→ Alert indication
 - Alerts sent to installer or homeowner
- Rules to reduce false positives



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SmRF Viewer

SmRF Viewer

Last refresh: 5:56:26 PM

Active Smart Radon Fan devices (5)

Viewer is collapsing 159 duplicate table entries into their newest matching SRF row.

Site	Device	Daily	7-Day Avg	Fan	Status	Reason	Last Report	Actions
Gutmann Merged 60 identifiers	radonbox-b3df michaeljgutmann@gmail.com	1.8 SAFE	1.8 SAFE	OFF	OK	ST_OFF_STREAK_2	06/13/2026, 1:00 PM	Card Log Chart Disable Reports Delete
Brinks Merged 51 identifiers	radonbox-dba4 michaeljgutmann@gmail.com	2.2 ELEVATED	4.4 HIGH	ON	OK	HOLD_STATE	06/13/2026, 1:00 PM	Card Log Chart Disable Reports Delete
Billie Merged 6 identifiers	radonbox-fb3f michaeljgutmann@gmail.com	3.2 ELEVATED	N/A NOT READY	ON	ERROR	HOLD_STATE	06/13/2026, 1:00 PM	Card Log Chart Disable Reports Delete
Eastman Merged 45 identifiers	radonbox-dbef	1.9 SAFE	4.1 HIGH	OFF	OK	ST_OFF_STREAK_2	06/13/2026, 1:00 PM	Card Log Chart Delete
Gutmann Merged 2 identifiers	radonbox-2b45	1.2 SAFE	N/A NOT READY	ON	OK	INSTALLATION_DAY_FORCE	06/12/2026, 1:00 PM	Card Log Chart Delete



Smart Radon Fan

The household loses power

- The radon fan, if active, stops as before
- When power is restored, SmRF will reboot and return to its pre-interrupted state
- The SmRF Back Office (SmRF Viewer) reports the installation as offline



Smart Radon Fan

The SmRF Raspberry Pi fails

- SmRF Viewer reports the installation as offline



Smart Radon Fan

SmRF loses Internet connection

- SmRF Viewer reports the installation as offline
- Locally, the SmRF continues to run, and may continue normally
- SmRF regularly and repeatedly attempts re-connection
- When connectivity is restored, SmRF re-connects automatically



Smart Radon Fan

Home WiFi network fails

- SmRF Viewer reports the installation as offline
- When connectivity is restored, SmRF re-connects automatically



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Home WiFi router changes device IPs

- New SmRF Raspberry Pi IP is available from the smartphone AP interface
- SmRF Viewer is updated with the new IP
- The Shelly Smart Plug is auto-rediscovered by SmRF
- Locally, the SmRF continues to run normally



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Airthings CRM can't be read (or produces questionable readings)

- Radon fan is turned ON as default
- SmRF UI shows failure
- SmRF Viewer is informed



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Shelly Smart Plug can't be accessed

- SmRF UI shows failure
- SmRF Viewer is informed



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Worst-case scenario

- SmRF Viewer reports the installation as offline
- The Smart Plug is removed, and the fan is plugged directly into the fan receptacle



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Fault-tolerant Testing

- Development testing of identified failure modes
- In-field Beta test results



Smart Radon Fan

SmRF Diagnostic AI Agent

- Available at smartradonfan.com
- Leads homeowner/installer through diagnostics and next steps
- Under development (May 2026)



Smart Radon Fan



Let **SmRF**
protect your home



Know Your
System is Working



Automated
Radon Mitigation



Real-Time
Feedback & Control



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Thanks for taking a look at SmRF!